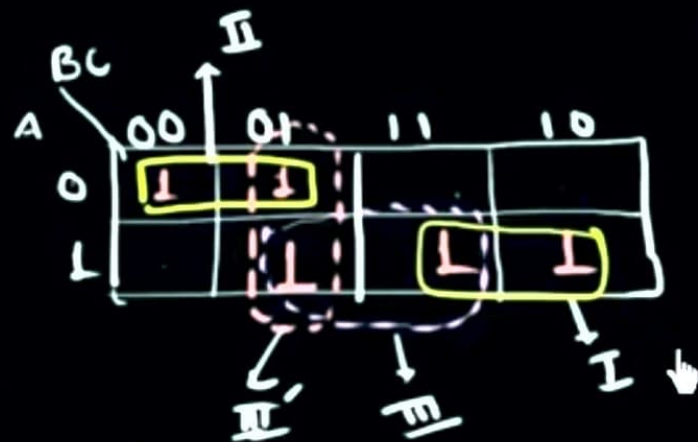


Ex:- $F(A, B, C) = \sum m(0, 1, 5, 6, 7)$



$$F = I + II + III$$

$$F = AB + \bar{A}\bar{B} + AC$$

NOTE

* The result is minimum
but ~~can~~ not be same
may
or unique

$$F = I + II + III'$$

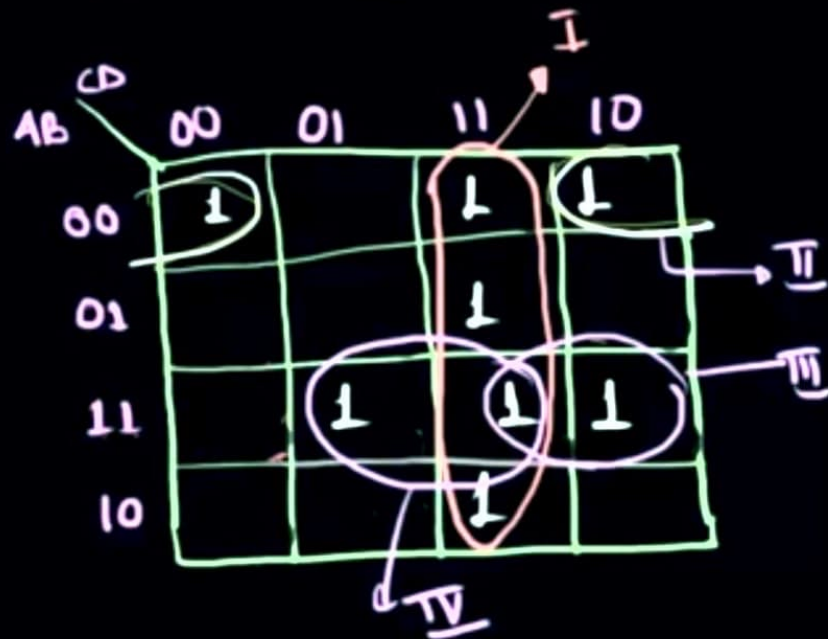
$$F = AB + \bar{A}\bar{B} + \bar{B}C$$

Ex: -i)

$$f(A, B, C, D) = \sum m(1, 3, 4, 5, 6, 7)$$

$\downarrow$ MSB $\downarrow$ LSB

Q1 :-



$$F = \overline{I} + \overline{II} + \overline{III} + \overline{IV}$$

$$F = CD + \bar{A}\bar{B}\bar{D} + ABC + ABD$$

$$m_2 = 1$$

$$m_3 = 1$$

$$m_7 = 1$$

$$m_{11} = 1$$

$$m_{13} = 1$$

$$m_{14} = 1$$

$$m_{15} = 1$$

✓ combine

—n—

—11—

—11—

2 1's $\Rightarrow$ 1 literal
if reduced

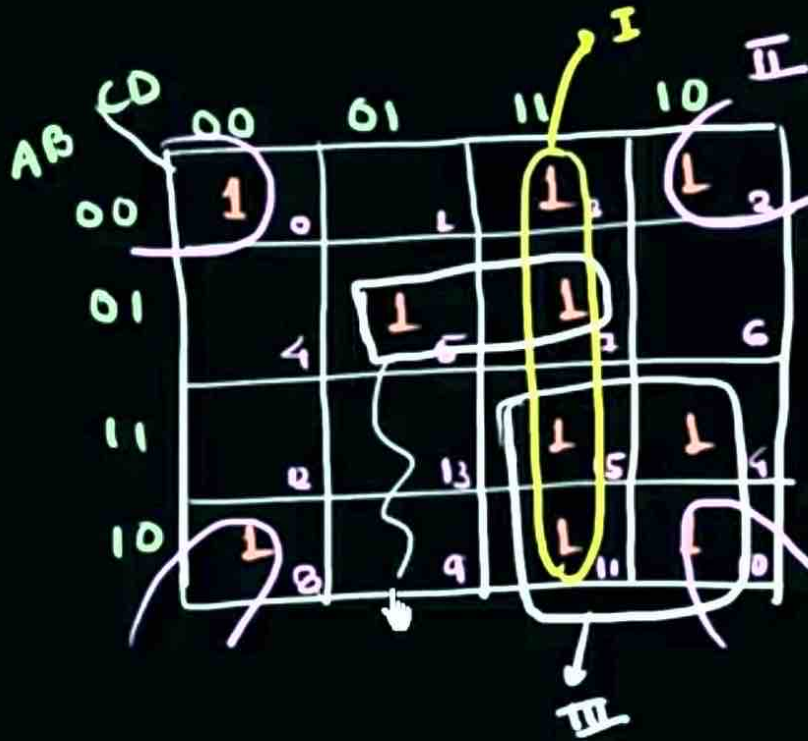
4 1's $\Rightarrow$ 2 —n—

8 1's $\Rightarrow$ 3 ip —r—

16 1's $\Rightarrow$ 4 —r—

Ex :- ii) $f(A, B, C, D) = \sum m(0, 2, 3, 5, 7, 8, 10, 11, 14, 15)$

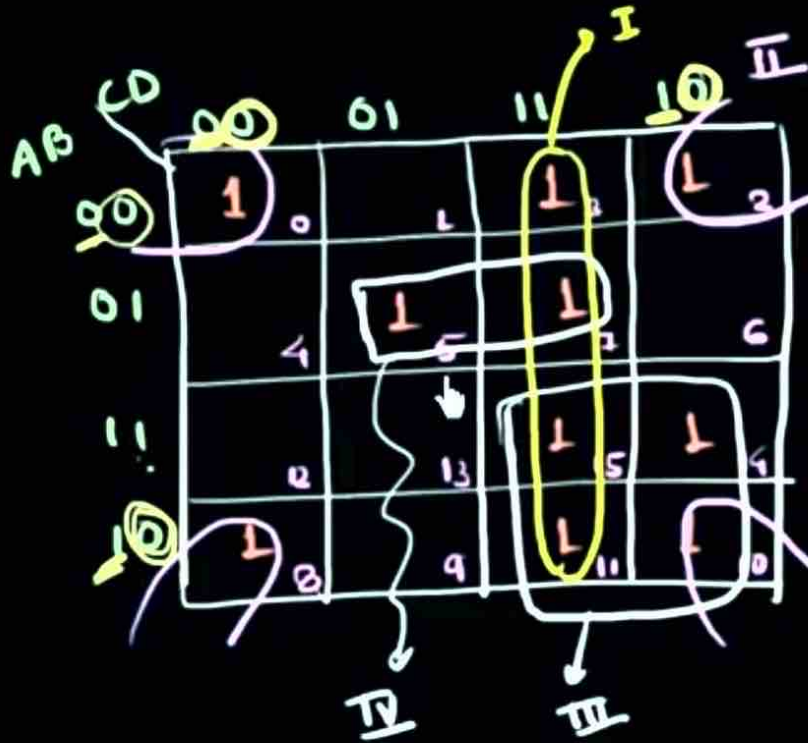
Sol :-



Ex:- ii)

$$F(A, B, C, D) = \sum m(0, 2, 3, 5, 7, 8, 10, 11, 14, 15)$$

Sol:-

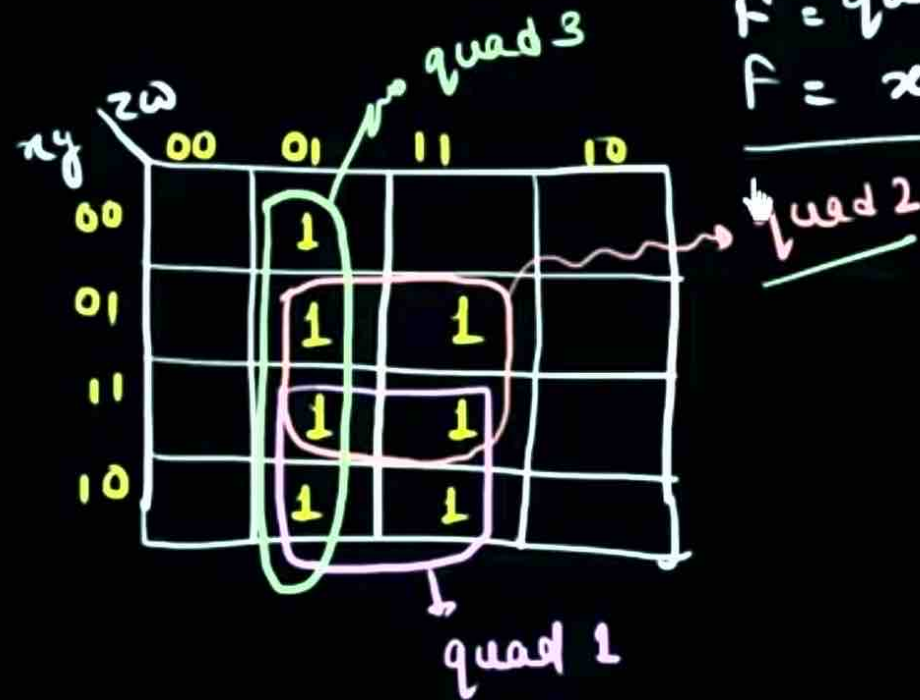


$$F = I + II + III + IV$$

$$F = CD + \bar{B}\bar{D} + AC + \bar{A}BD$$

4-Variable K'Map

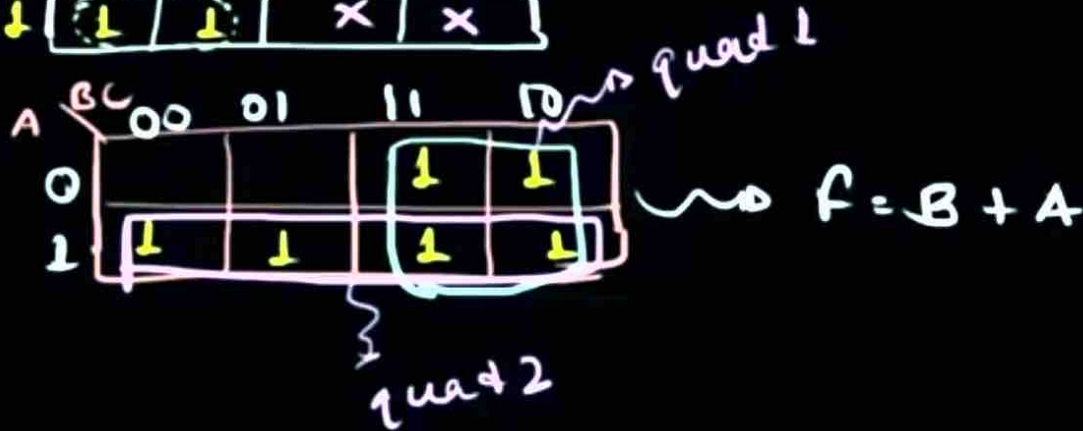
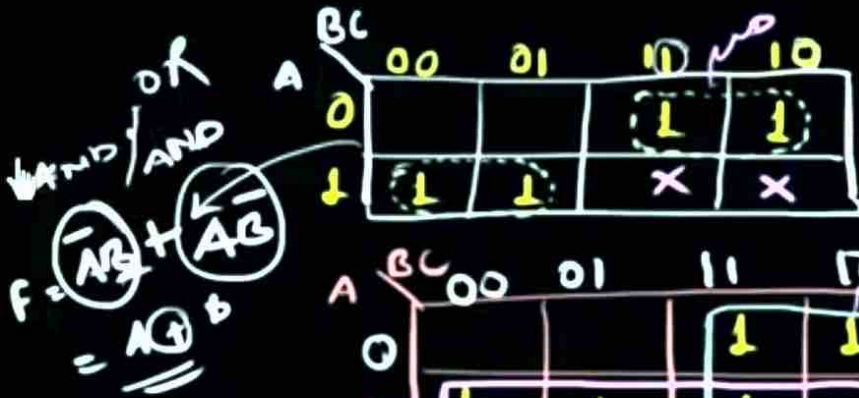
Ex:- $F(x, y, z, w) = \sum m(1, 5, 7, 9, 11, 13, 15)$



$$F = \text{quad 1} + \text{quad 2} + \text{quad 3}$$
$$F = xw + yw + \bar{z}w$$

Don't Case in K' Map

Ex:- $F(A, B, C) = \sum m(2, 3, 4, 5) + \sum d(6, 7)$



K' Map Using Max Term

Ex:-1

A

	BC	00	01	11	10
0		0	0	0	1
1		1	1	1	1

Group I: 00, 01, 11, 10 (Row 0)
Group II: 00, 01, 11, 10 (Row 1)

* *

$$F = 1 + 11$$

$$(F)' = (\overline{A}\overline{B} + \overline{A}C)'$$
 De Morgan's law -

$$F = (\overline{A}\overline{B})' \cdot (\overline{A}C)'$$

$$F = \underline{(A+B)} \cdot \underline{(A+C')}$$
 POS

K' Map Using Max Term

Ex:-1

BC

	00	01	11	10
0	0	0	0	1
1	1	1	1	1

A

Group I: (0,0), (0,1), (0,11) → $\bar{A}$

Group II: (0,0), (1,0), (1,1), (1,10) → $B\bar{C}$

$$F = A + B\bar{C}$$

* * *

$$F = 1 + 11$$

$$(\bar{F})' = (\bar{A}\bar{B} + \bar{A}C)'$$

De Morgan's law -

$$F = (\bar{A}\bar{B})' \cdot (\bar{A}C)'$$

$$\bar{F} = \underline{(A+B)} \cdot \underline{(A+C')}$$

POS

$$F = A + B\bar{C}$$

Ex: 2

$$F(A, B, C, D) = \sum m(1, 3, 4, 5, 9, 11, 14, 15)$$

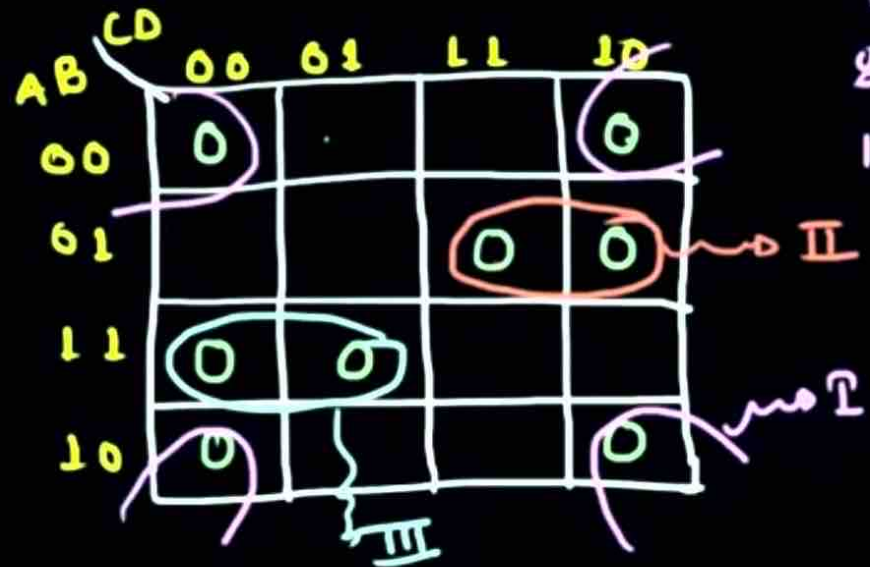
↳ min terms

Sol:-

$$F(A, B, C, D) = \prod M(0, 2, 6, 7, 8, 10, 12, 13)$$

↳ max terms

$$2^4 = 16$$



2
4
8
16

$$2^4 = 16$$

→ max terms

$$\overline{F} = \underline{\text{I}} + \underline{\text{II}} + \underline{\text{III}}$$

$$\overline{F} = \overline{B} \cdot \overline{D} + \underline{\underline{\overline{A} B C}} + A B \overline{C}$$

$$F = \underline{\underline{(B+D) \cdot (A+\overline{B}+\overline{C})(\overline{A}+\overline{B}+C)}}$$

	CD	00	01	11	10	
AB						
00	0				0	
01				0	0	II
11	0	0				
10	0				0	I

III

4
8
16

$\overline{A} \cdot B \cdot C$

$$F = (B \oplus D) \cdot (A + \overline{B} + \overline{C})(\overline{A} + \overline{B} + C)$$

A B C D
 0 - 0 0 0 0
 1 - 0 0 0 1
 3 - 0 0 1 1
 7 - 0 1 1 1
 8 - 1 0 0 0
 9 - 1 0 0 1
 11 - 1 0 1 1
 15 - 1 1 1 1

Step: - 1

Group	Minterm	Bin. Rep.			
		A	B	C	D
0	m_0	0	0	0	0
1	m_1	0	0	0	1
	m_8	1	0	0	0
2	m_3	0	0	1	1
	m_9	1	0	0	1
3	m_7	0	1	1	1
	m_{11}	1	0	1	1
4	m_{15}	1	1	1	1

Step: - 2

Quine-McCluskey Minimization Technique (Tabular Method)

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0
0
1
0
0
0
0
1
1

0 0 0 1
0 0 0 1

0 0 0 1
1 0 0 1

Step: -1

$n=4$
 $(n+1)=5$

Group	Minterm	Bin. Rep.			
		A	B	C	D
0	m_0	0	0	0	0
1	m_1	0	0	0	1
	m_2	1	0	0	0
2	m_3	0	0	1	1
	m_4	1	0	0	1
3	m_7	0	1	1	1
	m_{11}	1	0	1	1
4	m_{15}	1	1	1	1

Step: -2

Group	Matched pairs	Bin. Rep.			
		A	B	C	D
0	$m_0 - m_1$	0	0	0	-
	$m_0 - m_2$	-	0	0	0
1	$m_1 - m_3$	0	0	-	1
	$m_1 - m_4$	-	0	0	1
	$m_2 - m_4$	1	0	0	-
	$m_3 - m_7$	0	-	1	1
2	$m_3 - m_{11}$	-	0	1	1
	$m_4 - m_{11}$	1	0	-	1
	$m_7 - m_{15}$	-	1	1	1
	$m_{11} - m_{15}$	1	-	1	1

Quine-McCluskey Minimization Technique (Tabular Method)

Step:-2

000-
100-

Group	Matched pairs	Bin. Rep.			
		A	B	C	D
0	m_0-m_1	0	0	0	-
	m_0-m_8	-	0	0	0
	m_1-m_3	0	0	-	1
	m_1-m_9	-	0	0	1
	m_8-m_9	1	0	0	-
2	m_3-m_7	0	-	1	1
	m_3-m_{11}	-	0	1	1
	m_9-m_{11}	1	0	-	1
3	m_7-m_{15}	-	1	1	1
	$m_{11}-m_{15}$	1	-	1	1

Step:-3

Group	M.P.	B.R.			
		A	B	C	D
0	$m_0-m_1-m_8-m_9$	-	0	0	-
	$m_0-m_8-m_1-m_9$	-	0	0	-
	$m_1-m_3-m_9-m_{11}$	-	0	-	1
	$m_1-m_9-m_8-m_{11}$	-	0	-	1
2	$m_3-m_7-m_{11}-m_{15}$	-	-	1	1
	$m_3-m_{11}-m_7-m_{15}$	-	-	1	1

Step-3

Group	M.P.	B.R.				
		A	B	C	D	
0	$m_0 - m_1 - m_8 - m_9$	-	0	0	-	$\overline{B}\overline{C}$
	$m_0 - m_8 - m_1 - m_9$	-	0	0	-	
1	$m_1 - m_3 - m_9 - m_{11}$	-	0	-	1	$\overline{B}D$
	$m_1 - m_9 - m_3 - m_{11}$	-	0	-	1	
2	$m_3 - m_7 - m_{11} - m_{15}$	-	-	1	1	CD
	$m_3 - m_{11} - m_7 - m_{15}$	-	-	1	1	

P.I.

Quine-McCluskey Minimization Technique (Tabular Method)

$$Y = \bar{B}\bar{C} + CD$$

M.P.	B.R.				
	A	B	C	D	
$m_0 - m_0 - m_6 - m_9$	—	0	0	—	$\bar{B}\bar{C}$
$m_0 - m_8 - m_4 - m_9$	—	0	0	—	
$m_1 - m_3 - m_5 - m_{11}$	—	0	—	1	$\bar{B}D$
$m_1 - m_9 - m_5 - m_{11}$	—	0	—	1	

P.I.	Min terms involved	0	1	3	7	8	9	11	15
$\bar{B}\bar{C}$	0, 1, 8, 9	(X)	X			(X)	X		
$\bar{B}D$	1, 3, 9, 11		X	X			X	X	
CD	3, 7, 11, 15			X	(X)			X	(X)

AB \ CD	00	01	11	10
00	1	1	1	0
01	0	0	1	0
11	0	0	1	0
10	1	1	1	0

$$Y = CD + \bar{B}\bar{C} + \bar{B}D$$

$$Y = \bar{B}\bar{C} + CD$$

M.P.	B.R.				
	A	B	C	D	
$m_0 - m_0 - m_0 - m_0$	-	0	0	-	$\bar{B}\bar{C}$
$m_4 - m_8 - m_4 - m_8$	-	0	0	-	
$m_1 - m_3 - m_9 - m_{11}$	-	0	-	1	$\bar{B}D$
$m_5 - m_9 - m_5 - m_{11}$	-	0	-	1	

P.I.	min terms involved	0	1	3	7	8	9	11	15
$\bar{B}\bar{C}$	0, 1, 8, 9	(X)	X			(X)	X		
$\bar{B}D$	1, 3, 9, 11		X	X			X	X	
CD	3, 7, 11, 15			X	(X)			X	(X)

Comparison between Combinational & Sequential Ckt.

↓
O/p is only dependent
of present i/p

ex:- Adder

$$\begin{array}{r} 1 \\ + 0 \\ \hline 1 \\ \hline \end{array}$$

↓
 O/p depends on the
 present i/p as well
 as previous O/p & o/p's

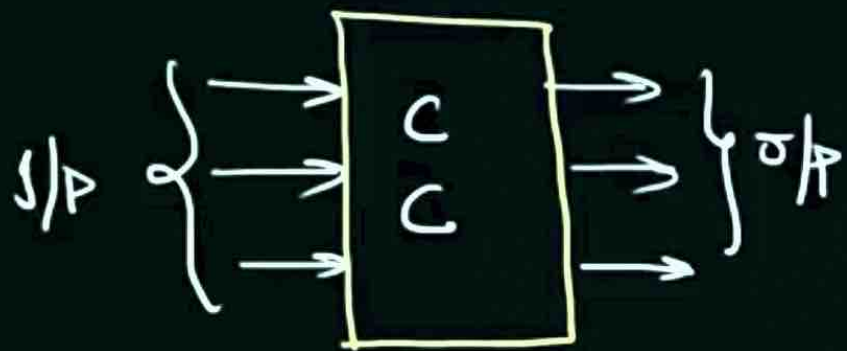
ex:- Counter

$$\begin{array}{r} + 1 \text{ prev O/P} \\ \hline \end{array}$$

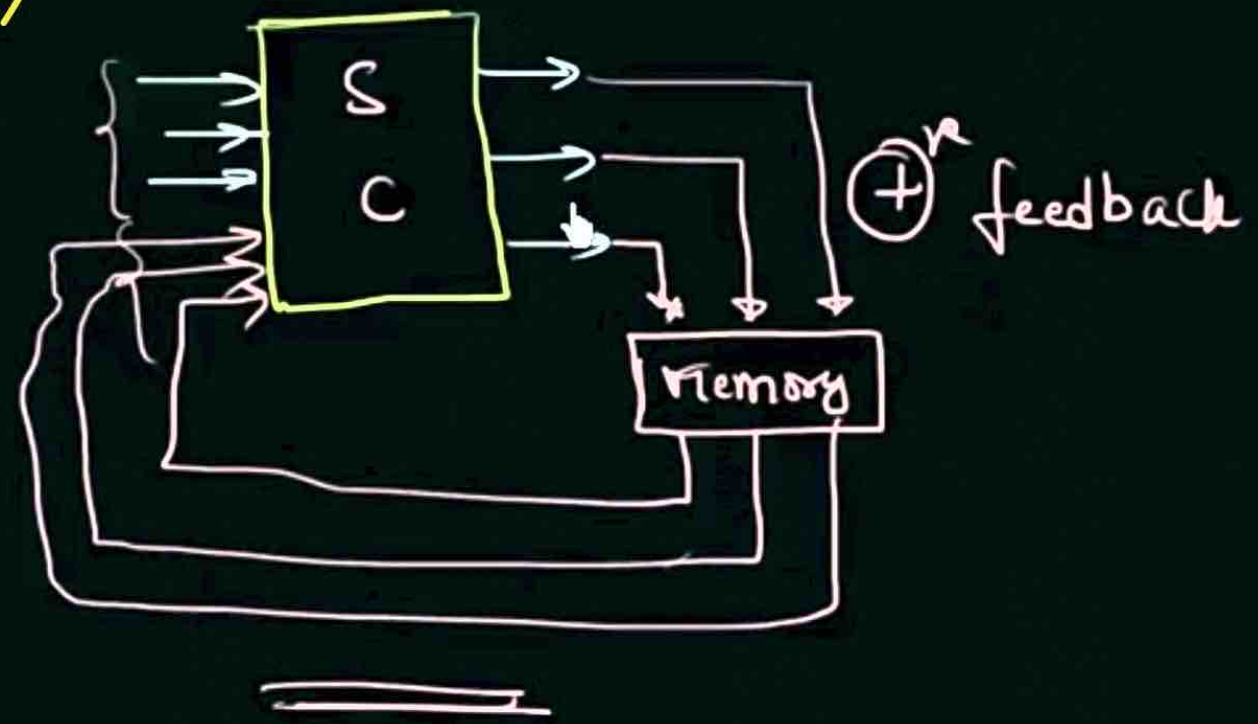
$$\begin{array}{l} 4+1 \rightarrow 5 \\ \hline 3+1 \rightarrow 4 \end{array} \quad \downarrow$$

...

sequential



combo



Half Adder [comb. ckt.]

* single bit no's

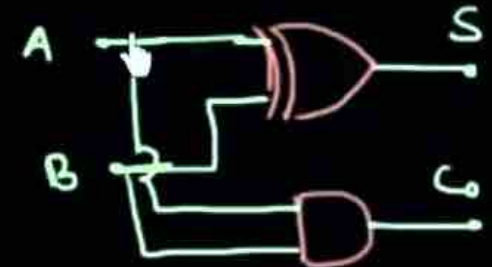
* it does not take
carry from previous sum



$$S = A \oplus B$$

$$C_o = A \cdot B$$

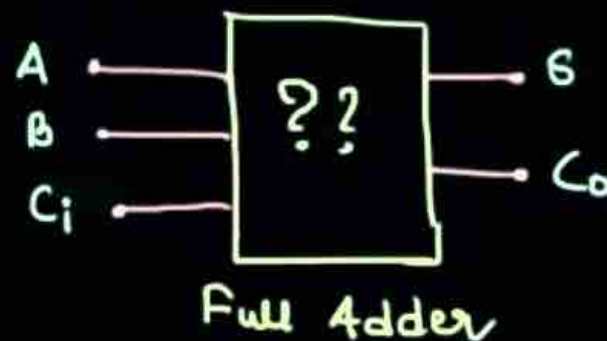
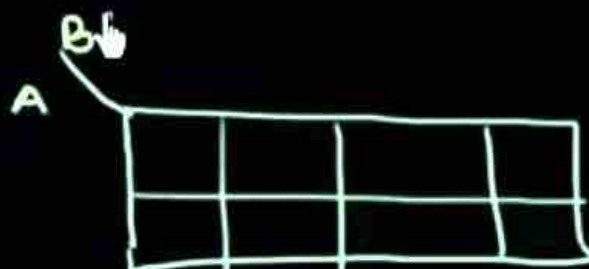
A	B	S	C _o
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



Half Adder

Full Adder [Part 2]

A	B	C _i	S	C _o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



$$r = 2$$

$$2 = \textcircled{2} - 0$$

$$\begin{array}{r} 1 \\ C_o + 1 \\ \hline S = 0 \end{array}$$

decimal (r=10)

$$\begin{array}{r} 1 \\ 13 \\ + 06 \\ \hline 1 \textcircled{2} \end{array}$$

$$\begin{array}{r|l} 1+1=0 & - \\ 0+0 & - \\ 0+1 & - \end{array} \begin{array}{l} 0 \\ 0 \\ 1 \end{array}$$

for C_0 -
Check board carry
 $S = A \oplus B \oplus C_i$

		$B C_i$			
		00	01	11	10
A	0	0	0	1	0
	1	0	1	1	1

$$C_0 = B C_i + A B + A C_i$$

$$\underline{\underline{C_0 = A B + C_i (A \oplus B)}}$$

$$\begin{array}{r} 10 + 13 = 23 \\ + 06 \\ \hline 1(3) \end{array}$$

$$S = A \oplus B \oplus C_i$$

		$B C_i$			
		00	01	11	10
A	0	0	0	1	0
	1	0	1	1	1

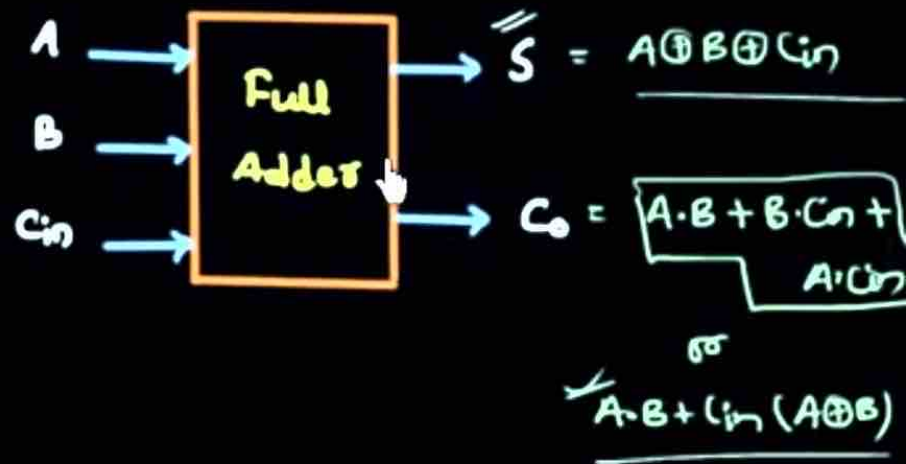
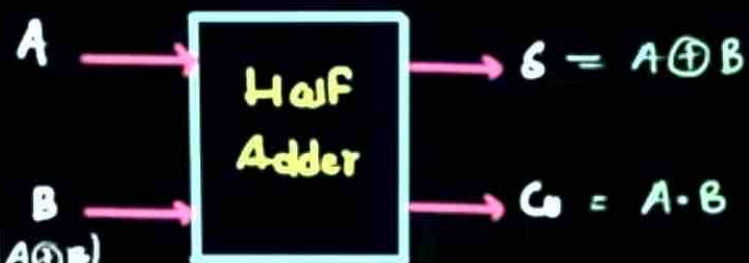
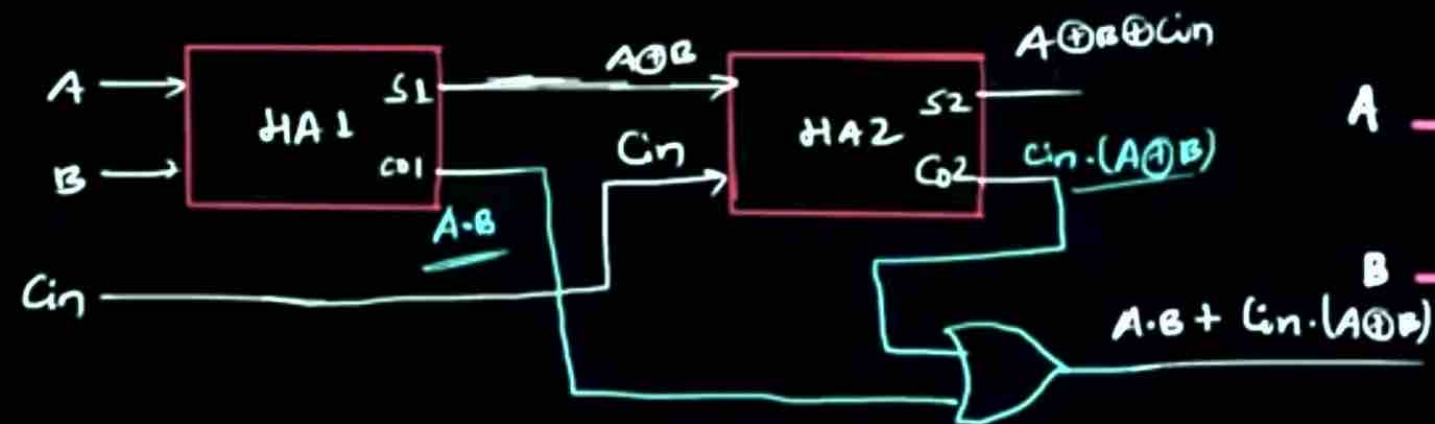
$$C_0 = A B + \bar{A} \bar{B} C_i + \bar{A} B C_i$$

$$= A B + C_i (\bar{A} \bar{B} + \bar{A} B)$$

$$= A B + C_i (A \oplus B)$$

Full Adder using Half Adder

$$S1 = A \oplus B$$



$$A \cdot B + C_{in}(A \oplus B) = A \cdot B + C_{in}(AB' + A'B)$$

$$\underline{X + \overset{1}{\cancel{X}} = X + 1}$$

$$= \underline{A \cdot B} + \underline{AB'}C_{in} + A'B C_{in}$$

$$= A(B + \overset{1}{\cancel{B}}C_{in}) + A'B C_{in}$$

$$= A(B + C_{in}) + A'B C_{in}$$

$$= A \cdot B + A \underline{C_{in}} + A'B \underline{C_{in}}$$

$$= A \cdot B + (A + A'B)C_{in}$$

$$= A \cdot B + (A + B)C_{in}$$

$$= A \cdot B + AC_{in} + BC_{in}$$

2:49 PM Tue 21 Feb

Today
2:49 PM

Edit



Comparison between Combinational & Sequential Ckt.

↓
O/p is only dependent
of present I/p

ex:-

Adder

$$\begin{array}{r} 1 \\ + 0 \\ \hline 1 \\ \hline \end{array}$$

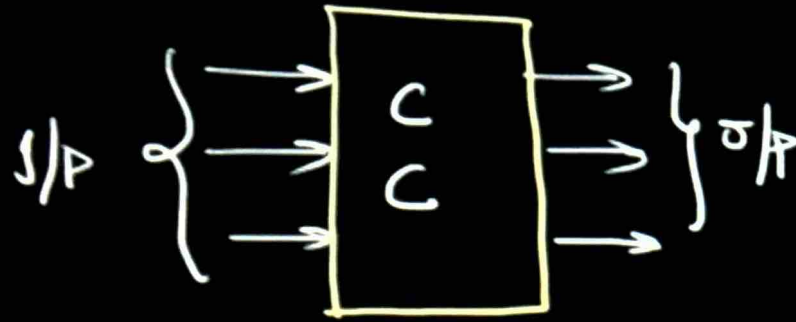
↓
O/p depends on the
present I/p as well
as previous O/p or I/p

ex:- Counter

+ 1 prev O/p

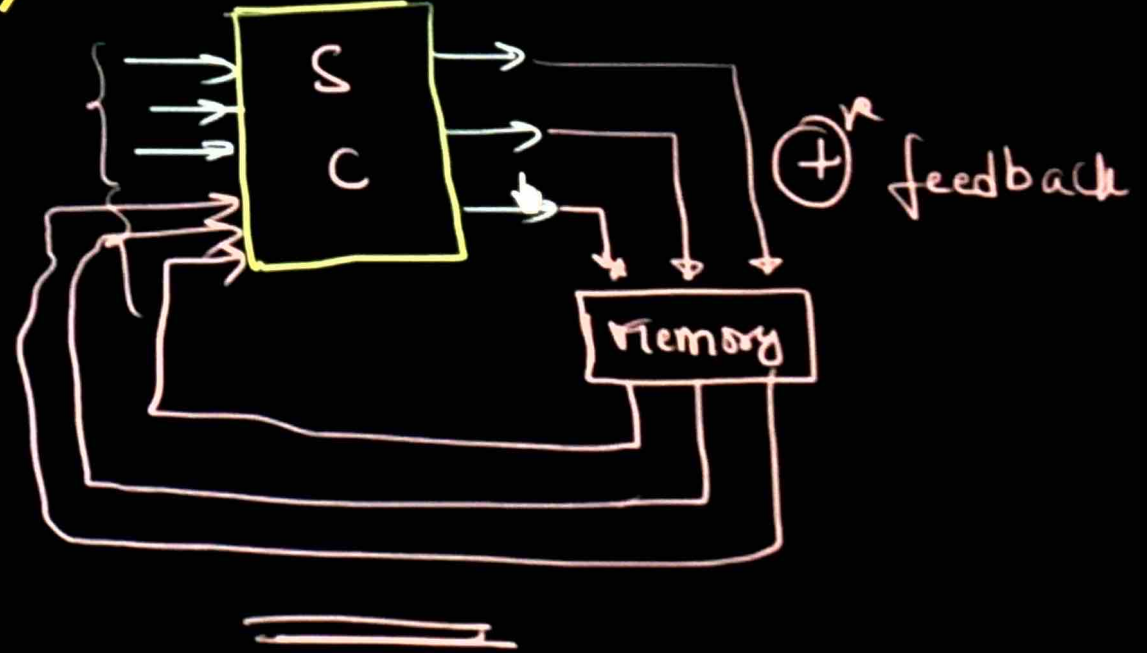
$$4 + 1 = 5$$

$$3 + 1 = 4$$

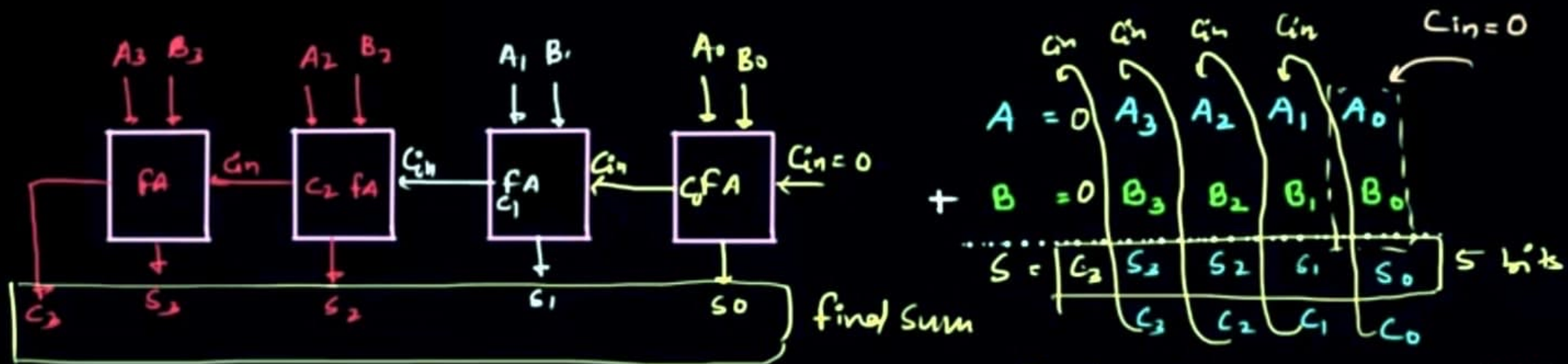


Ques.

→ sequential



Four Bit Parallel Adder using Full Adders



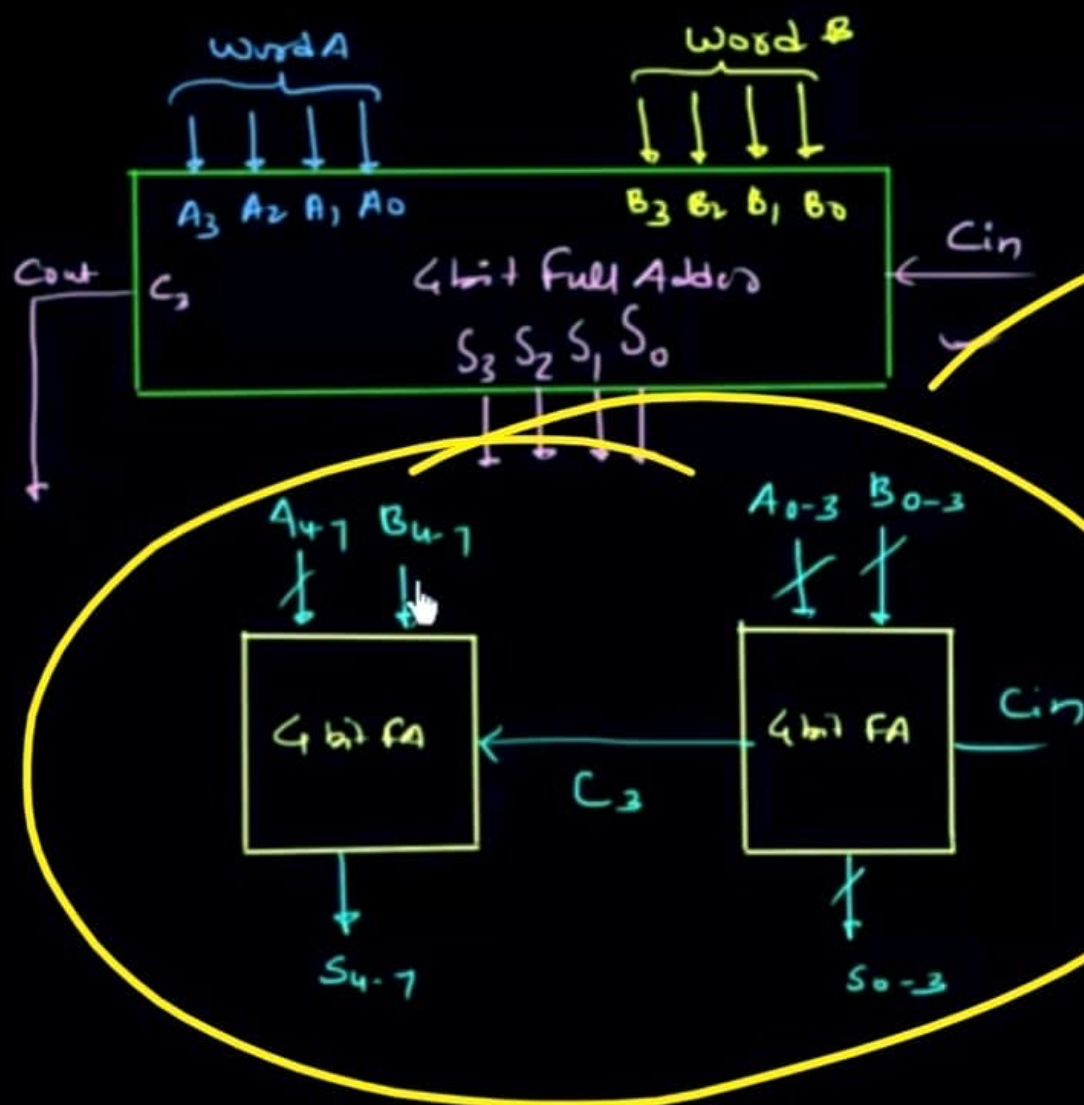
TC #1(283) $\Rightarrow$ IC for 4 bit full adder

$C_3 = 1$ or 0

4 bit sum
5 bit sum

4 bit
5 bit
5'
+ 5'
1101

4 Bit Parallel Adder using Full Adders



8 bit

FF

CA

$A = A_7 \dots A_0$

$B = B_7 \dots B_0$

$S = C_7, S_7 \dots S_0$

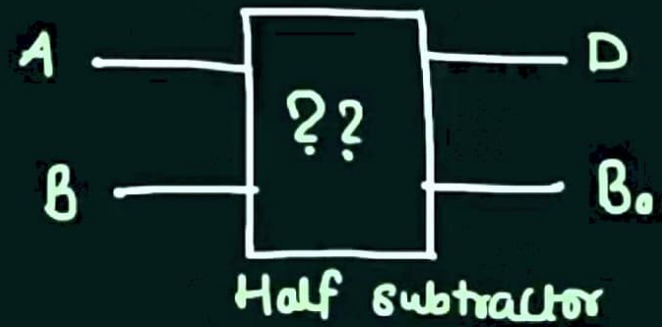
using 4 bit

4 Bit Parallel Adder using Full Adders

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HALF SUBTRACTOR

$$\begin{array}{r}
 \overset{2'}{0} \overset{2'}{0} \\
 \quad \quad \quad \curvearrowright \quad 2'=2 \\
 - \quad 0 \quad 1 \\
 \hline
 \quad 0 \quad 1 \quad A_0 \\
 \hline
 \end{array}$$



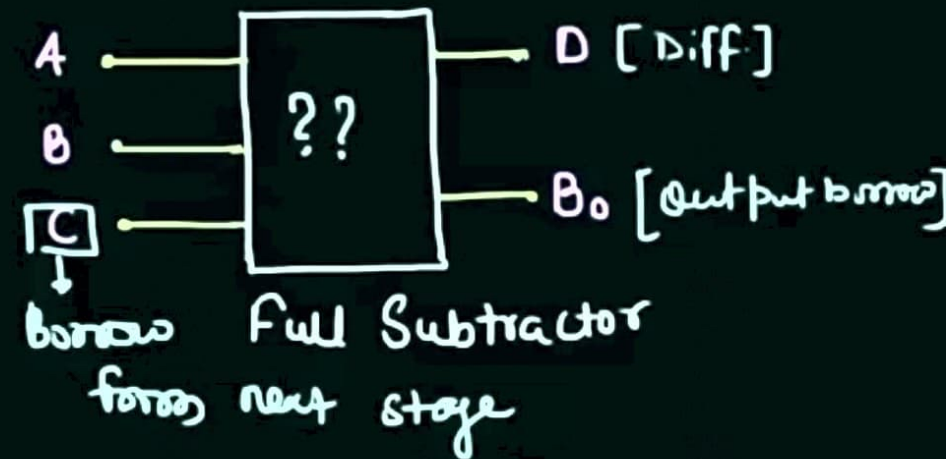
A	B	D	B ₀
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$D = A \oplus B$$

$$B_0 = \bar{A} B$$

FULL SUBTRACTOR

A	B	C	D	B ₀
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1



$0 - 1 = 0$	1	1
$0 - 1 = 1$	0	1
$1 - 0 = 1 - 0 = 1$	1	0
$1 - 0 = 1 - 1 = 0$	0	0
$1 - 1 = 0 - 0 = 0$	0	0
$1 - 1 = 0 - 1$	1	1

Borrow Full Subtractor
from next stage

for D :-

	BC	00	01	11	10
A	0	0	1	0	1
1	1	1	0	1	0

imp

Check borrow
conf

$$D = A \oplus B \oplus C$$

Full adder
Parity gen.

for B_0 -

	BC	00	01	11	10
A	0	0	1	1	1
1	1	0	0	1	0

$$B_0 = BC + \bar{A}C + \bar{A}B$$

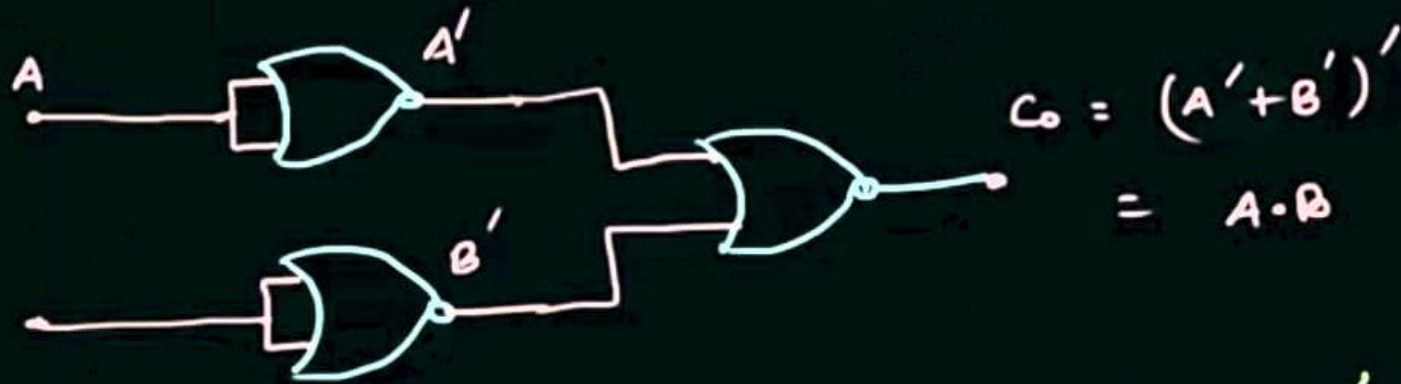
Full Subtractor | Easy Explanation

Half Adder using NOR Gate Only

$$S = A \oplus B$$

$$C_0 = \underline{A \cdot B}$$

$$\begin{matrix} A' \\ B' \end{matrix} \rightarrow \text{NOR Gate} \rightarrow (A+B)' = \boxed{A'} B' = A \cdot B$$

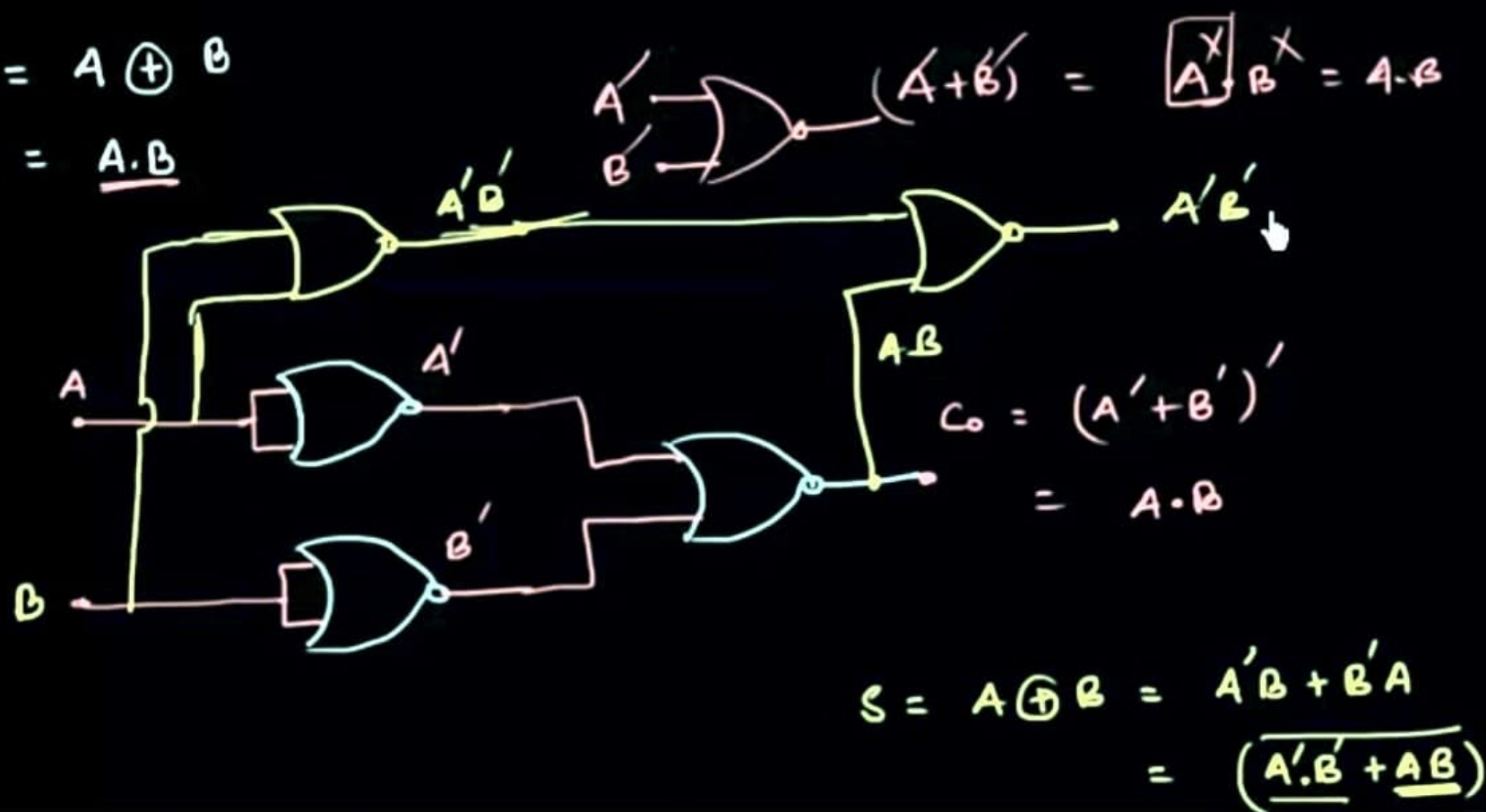


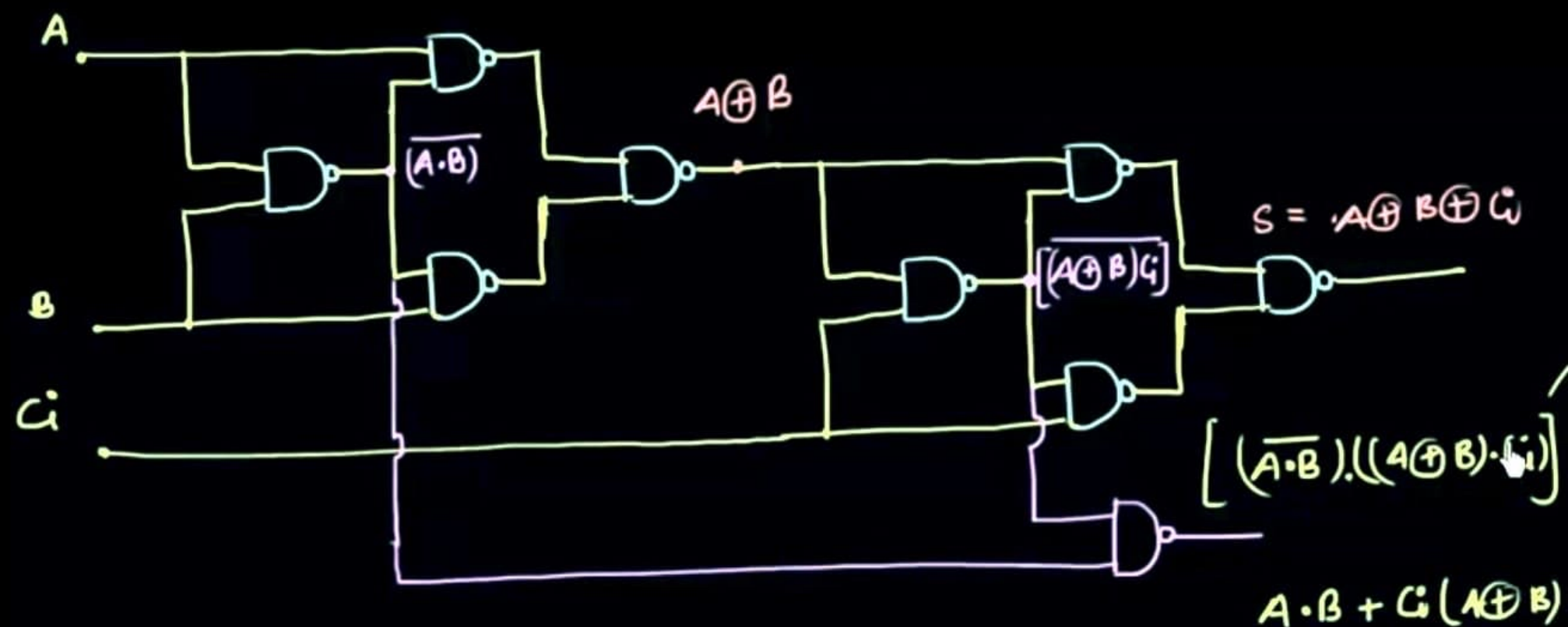
$$S = A \oplus B = A'$$

Half Adder using NOR Gate Only

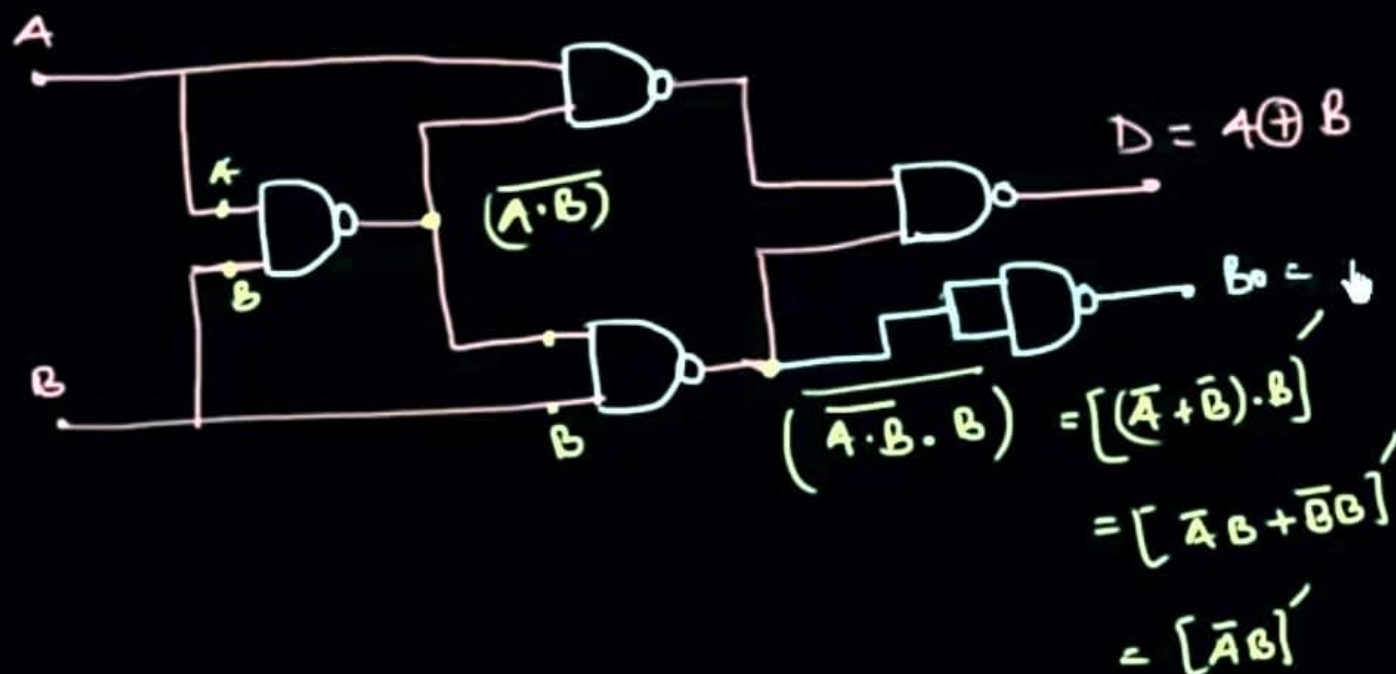
$$S = A \oplus B$$

$$C_0 = \underline{A \cdot B}$$





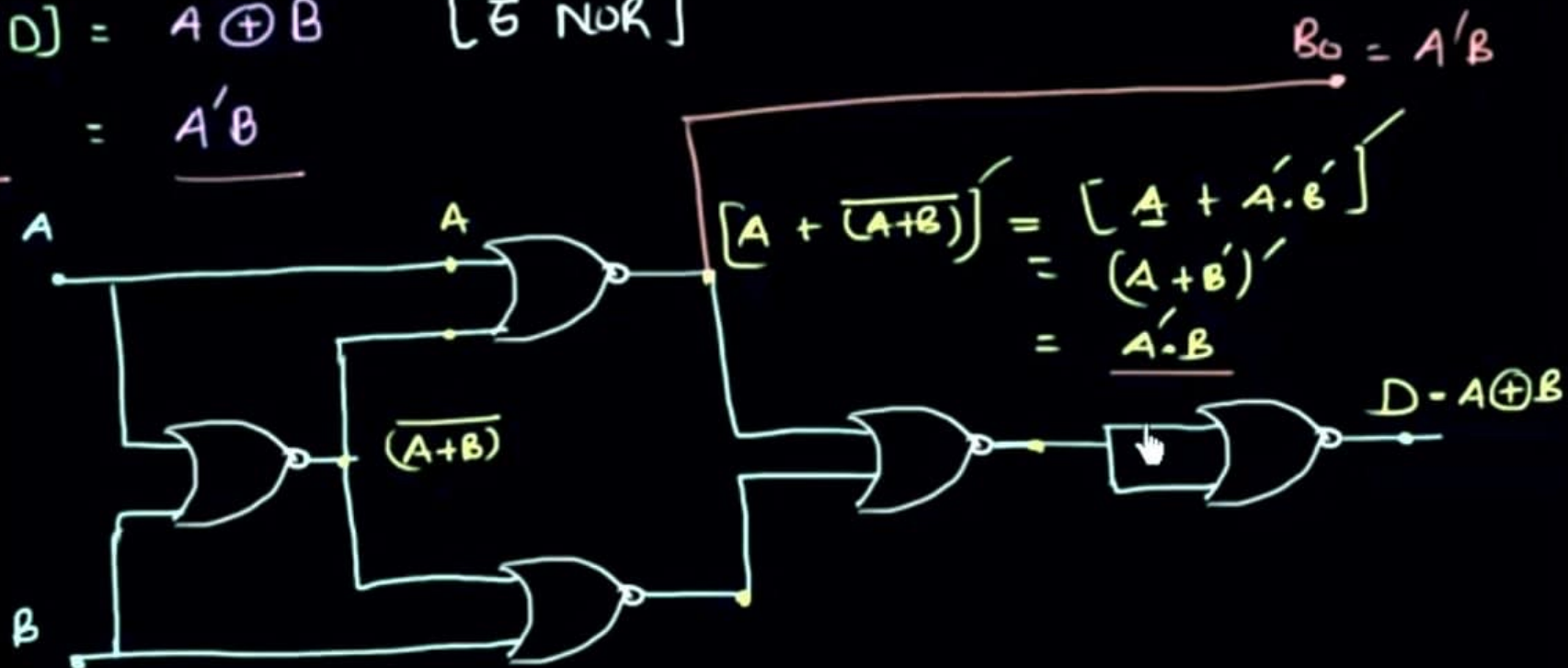
0	1	1	1
1	0	1	0
1	1	0	0



Realizing Half Subtractor using NOR gate only

$$\text{Difference } [D] = A \oplus B \quad [5 \text{ NOR}]$$

$$\text{Borrow } [B_0] = \underline{A'B}$$



Full Subtractor using NAND Gates Only

Minuend
msb
A

Subtrahend
lsb
B

Comb. ckt.

Carry in
 B_i

Diff
 D

Borrow out
 B_o

A	B	B_i	D	B_o
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

O/p or o/p's $\rightarrow$ present i/p or i/p's

$$D = A - B - B_i$$

$$D = 0 - 0 - 1$$

$$= 0 - 1$$

$$= 1$$

$$S = A \oplus B \oplus C_i$$

$$D = A \oplus B \oplus B_i$$

Karnaugh Map for D and B_o

A	$B B_i$			
	00	01	11	10
0	0	1	1	1
1	0	0	1	0

Groupings for D (I, II, III) and B_o (I, II, III)

$$B_o = \bar{A}B + \bar{A}\bar{B}B_i +$$

$$AB B_i$$

$$B_o = I + II + III$$

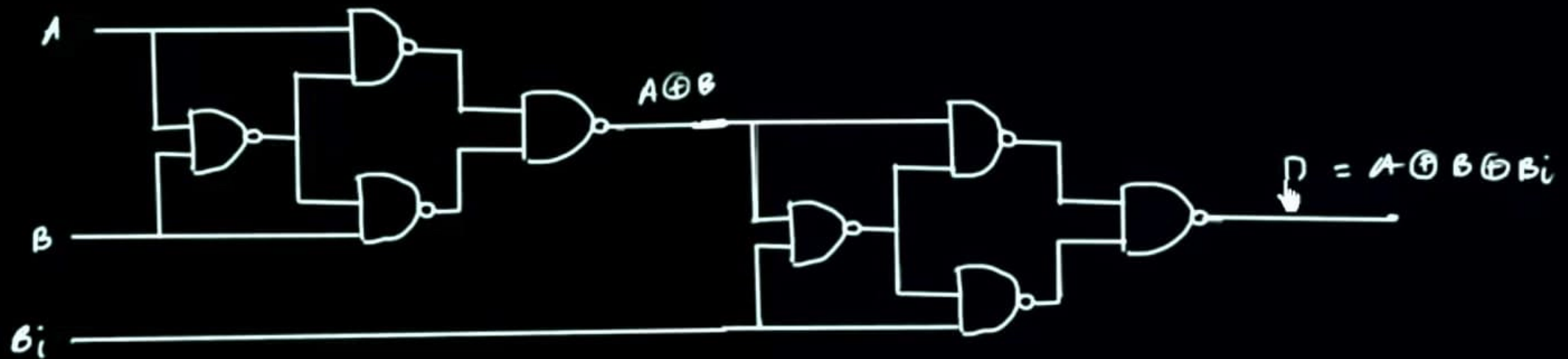
$$B_o = \bar{A}B + \bar{A}B_i + B B_i$$

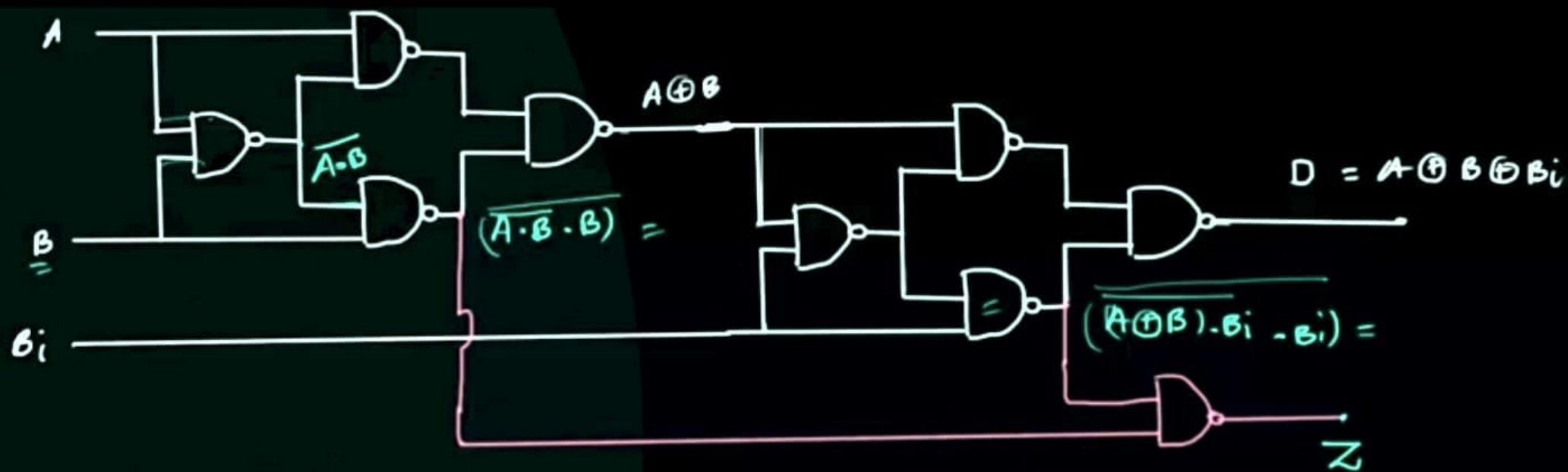
Realizing Full Subtractor using NAND Gates only (Part 1)

$$B_0 = \overline{A}B + \overline{A}\overline{B}B_i + AB\overline{B}_i$$

$$B_0 = \text{I} + \text{II} + \text{III}$$

$$\underline{B_0} = \overline{A}B + \overline{A}B_i + B\overline{B}_i$$





$$\overline{A \cdot B \cdot B_i} = \overline{AB} + \overline{B}$$

$$AB + \overline{B}$$

$$\underline{\underline{A + \overline{B}}}$$

$$Z = \overline{(A + \overline{B}) \cdot (A \oplus B + \overline{B_i})}$$

$$\overline{(A \oplus B) \cdot B_i - B_i} = \overline{(A \oplus B) \cdot B_i} + \overline{B_i}$$

$$= (A \oplus B) \cdot B_i + \overline{B_i}$$

$$= \underline{\underline{A \oplus B + \overline{B_i}}}$$

$$\overline{\overline{A \cdot B} \cdot B} = \overline{\overline{AB} + \overline{B}}$$

$$AB + \overline{B}$$

$$\underline{\underline{A + \overline{B}}}$$

$$Z = \overline{(A + \overline{B}) \cdot (A \oplus B + \overline{B_i})}$$

$$\overline{(A \oplus B) \cdot \overline{B_i} \cdot B_i} = \overline{(A \oplus B) \cdot \overline{B_i}} + \overline{B_i}$$

$$= (A \oplus B) \cdot B_i + \overline{B_i}$$

$$= \underline{\underline{A \oplus B + \overline{B_i}}}$$

$$\overline{XOR} = XNOR$$

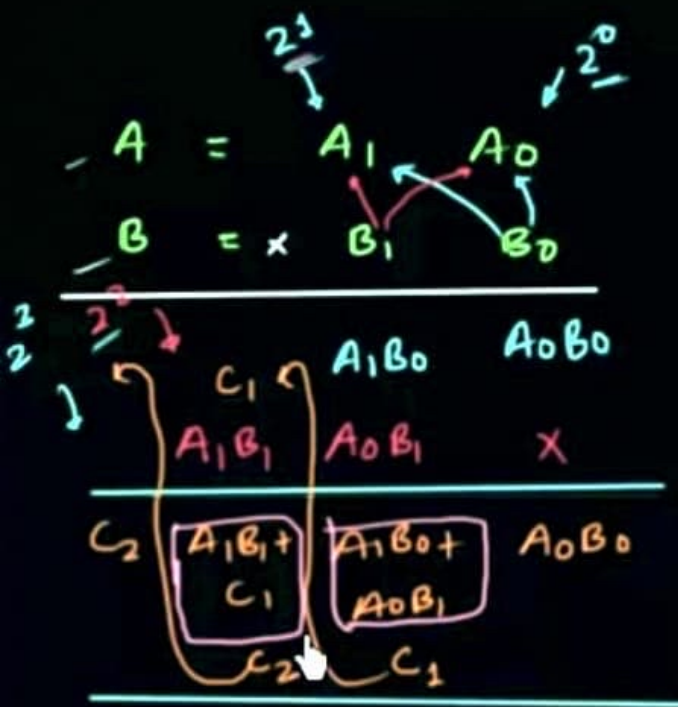
$$= \overline{(A + \overline{B})} + \overline{(A \oplus B + \overline{B_i})}$$

$$= \overline{A} \cdot B + \overline{A \oplus B} \cdot B_i$$

$$= \overline{A} \cdot B + A \odot B \cdot B_i$$

$$= \overline{A} B + (AB + \overline{A} \overline{B}) \cdot B_i = \overline{A} B + AB B_i + \overline{A} \overline{B} B_i$$

2-Bit Multiplier



base = 2

$$A_0 \times 2^0 \times B_1 \times 2^1$$

$$(A_0 B_1) \times 2^1$$

$$A_1 \times 2^1 \times B_1 \times 2^1$$

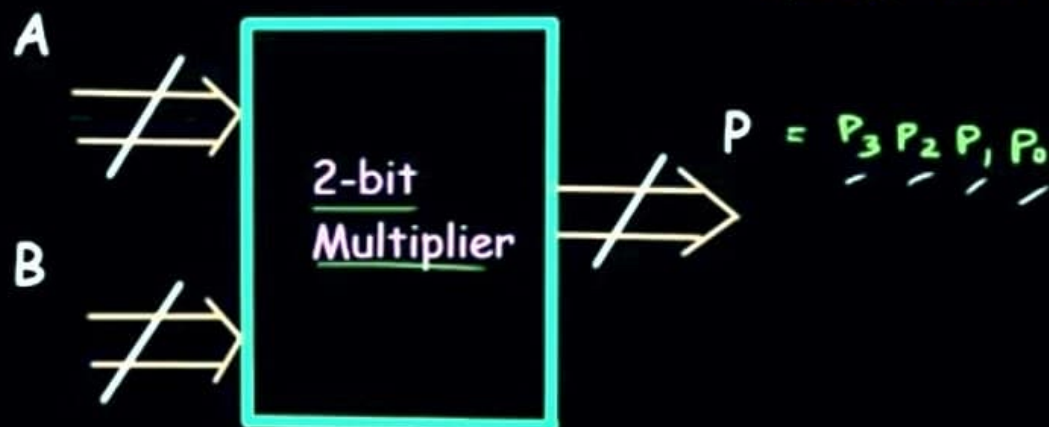
$$A_1 B_1 \times 2^2$$

$$P_0 = A_0 B_0$$

$$P_1 = A_1 B_0 + A_0 B_1$$

$$P_2 = A_1 B_1 + C_1$$

$$P_3 = C_2$$



2-Bit Multiplier Using Half Adders

$$P_0 = A_0 B_0$$

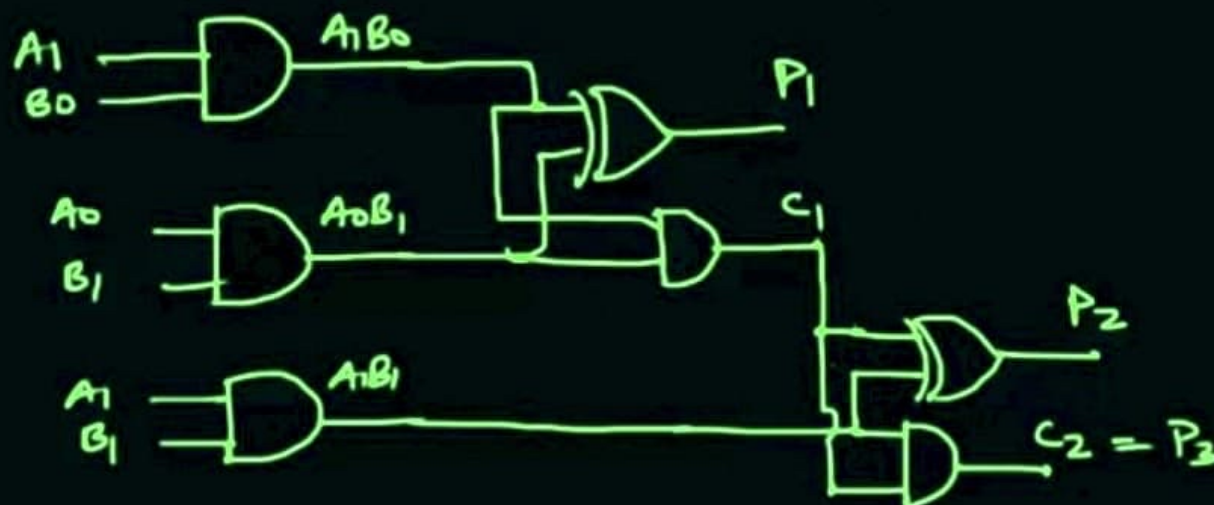
$$P_1 = \underbrace{A_1 B_0 + A_0 B_1}_{H.A.}$$

$$A_1 B_1 \times 2^2$$

$$P_2 = \underbrace{A_1 B_1 + C_1}_{H.A.}$$

$$P_3 = \underline{\underline{C_2}}$$

$$\begin{aligned} \text{Sum} &= X \oplus Y \\ \text{Carry} &= X \cdot Y \end{aligned}$$



$$\begin{array}{r} A = 11 (3) \\ B = \times 10 (2) \\ \hline 00 \\ 11 \times \\ \hline 110 (6) \end{array}$$

2-Bit Multiplier Using Half Adders

$$P_0 = A_0 B_0$$

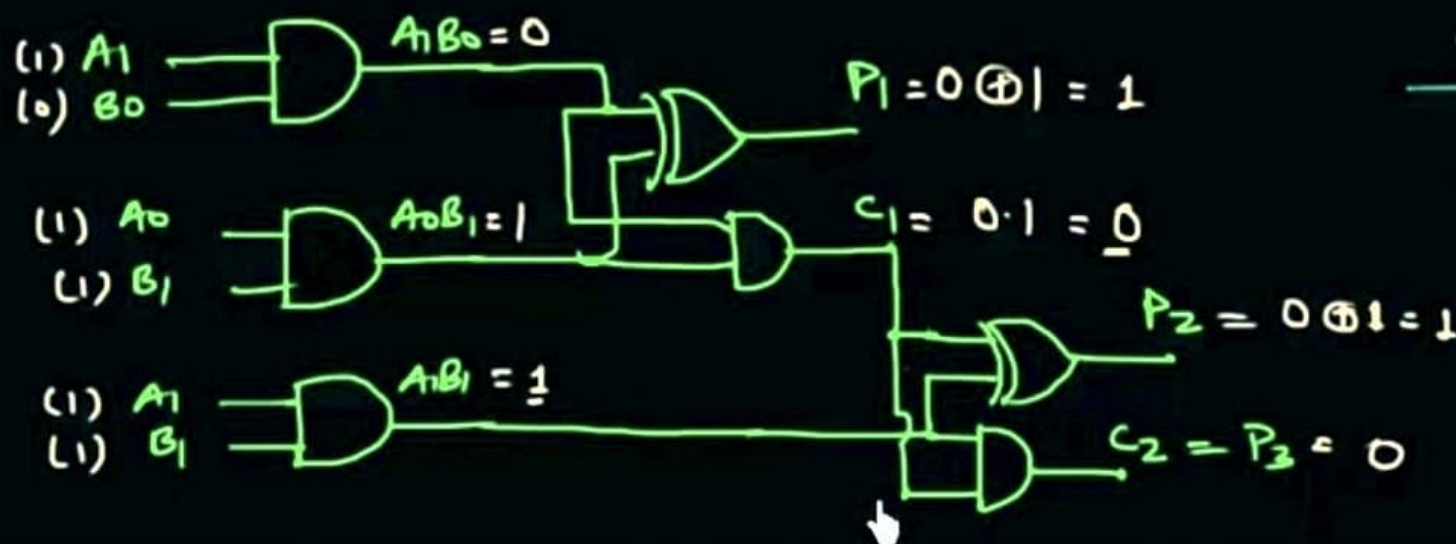
$$P_1 = \underbrace{A_1 B_0 + A_0 B_1}_{H.A.}$$

$$A_1 B_1 \times 2^2$$

$$P_2 = \underbrace{A_1 B_1 + C_1}_{H.A.}$$

$$P_3 = C_2$$

$$\begin{array}{r} A_1 B_0 \oplus A_0 B_1 \\ \text{Sum} = X \oplus Y \\ \hline \text{Carry} = X \cdot Y \end{array}$$



$$\begin{array}{r} A = 1 \ 1 \ (3) \\ B = \times \ 1 \ 0 \ (2) \\ \hline 0 \ 0 \\ 1 \ 1 \times \\ \hline 1 \ 1 \ 0 \ (6) \end{array}$$

$\uparrow P_2 \quad \uparrow P_1 \quad \uparrow P_0$